

ChartJunkGPT: Can GPT-4.1 Interpret Visually Embellished Charts?



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Large language models can read basic data visualizations¹ – but what about embellished charts that contain “chart junk”?

Introduction

In this work, we evaluate the ability of the GPT-4.1 language model to answer basic questions about embellished visualizations, comparing responses on embellished charts to more plain counterparts. We employ similar questions and the same 14 pairs of embellished and plain charts as a prior study with human subjects² with chart images from a separate replication study³.

Results

Embellished Charts

GPT always understands the **Subject**

“The chart [...] breaks down ‘**The Cosmetics Dollar**’ to show **where the money goes**, dividing it between the manufacturer and the retailer.”

GPT can interpret subtle **Value Messages**

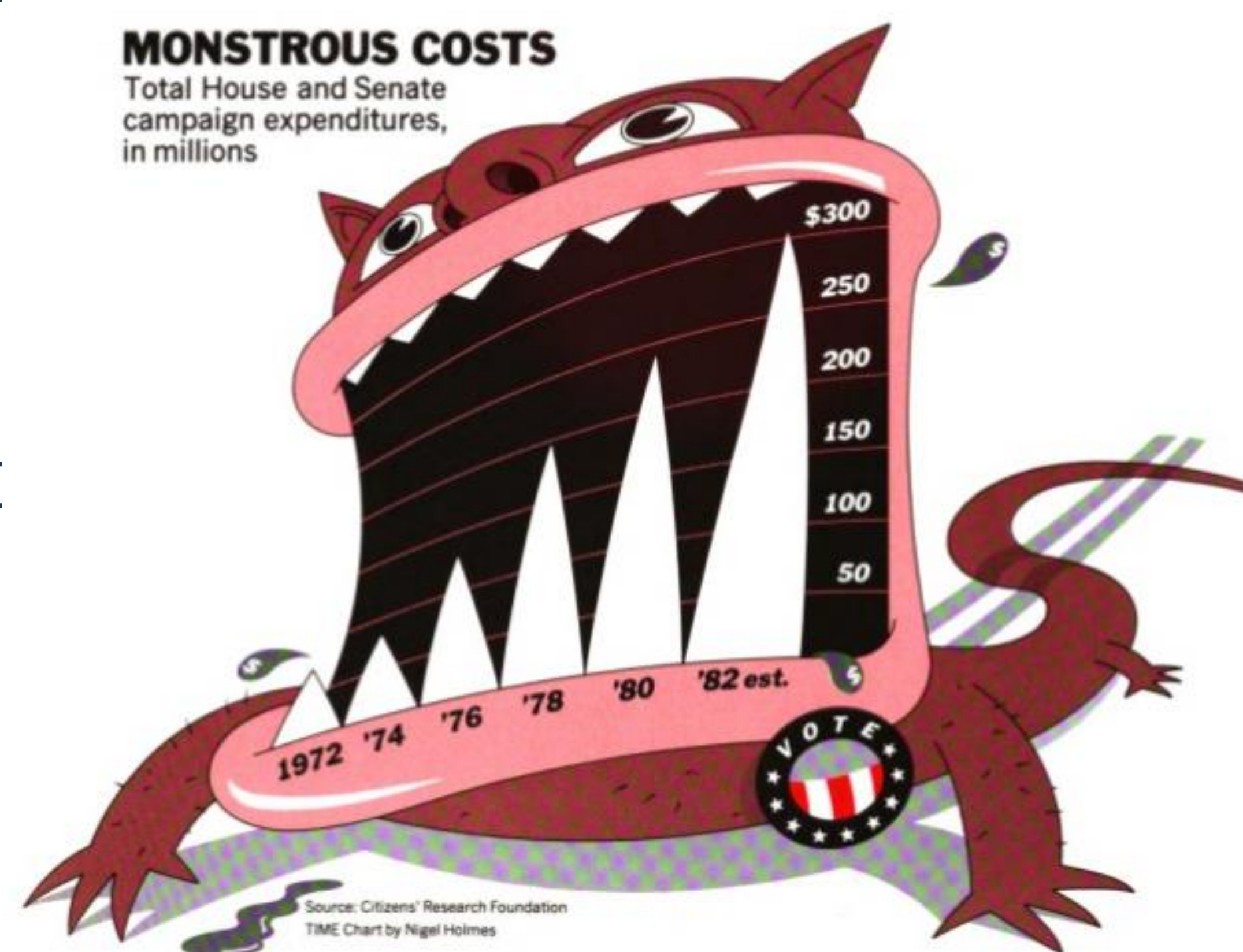
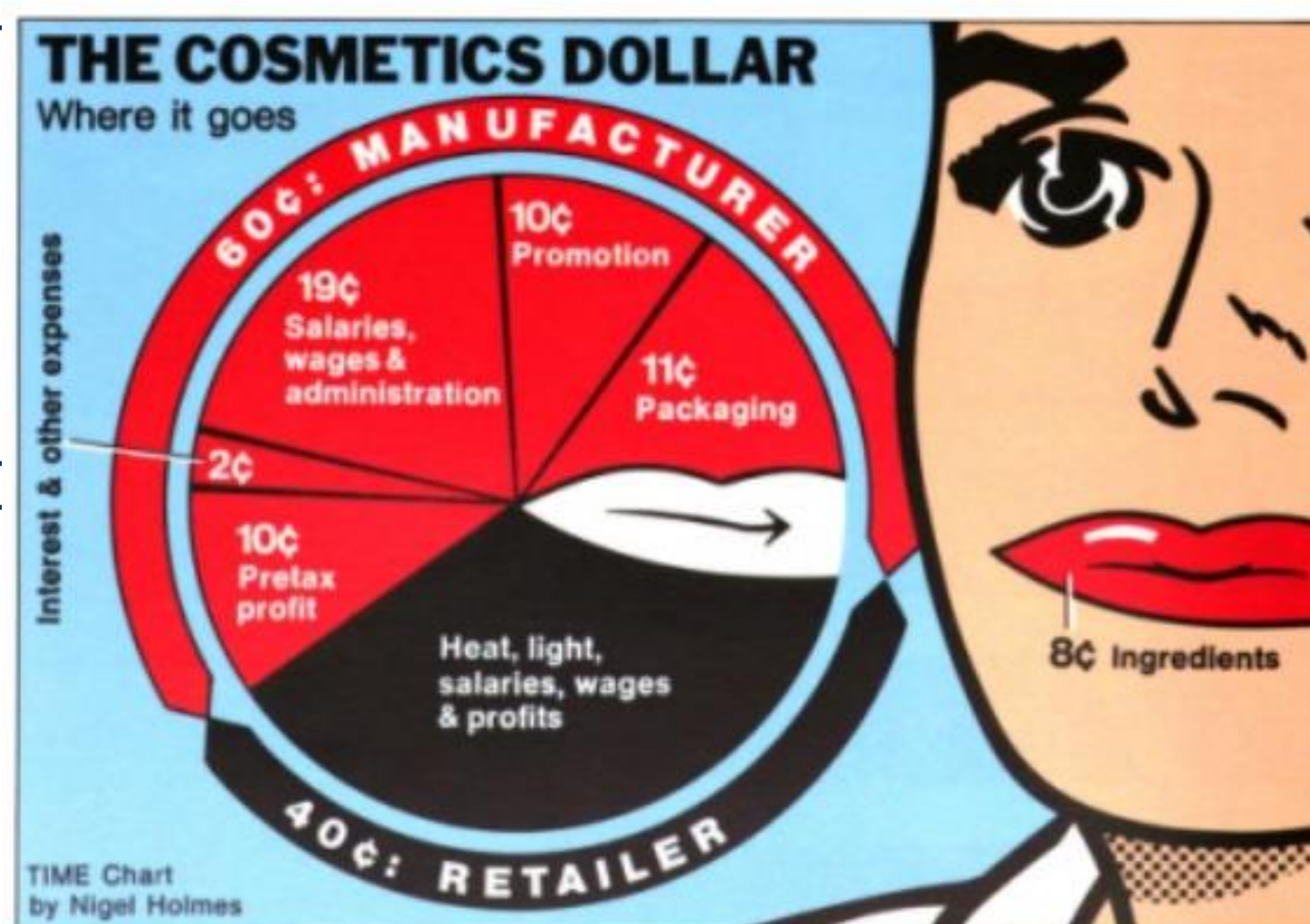
“The use of a **large, stylized face** with bright red lips **draws attention to the idea** of cosmetics and their perceived value.”

GPT mostly understands **Values & Trends**, but distorted axes can pose a challenge

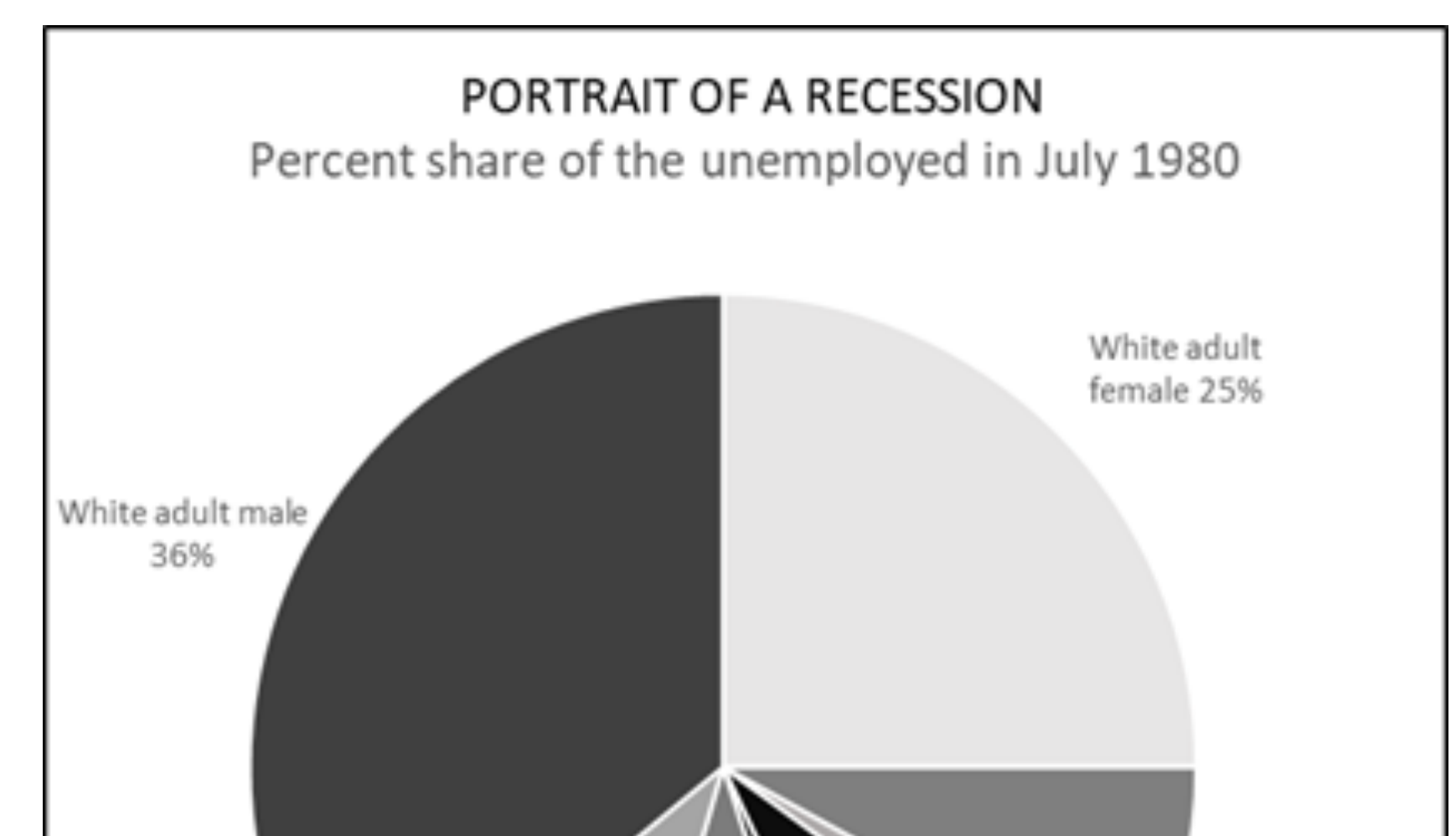
“The **basic trend** is an **increase** in total House and Senate campaign expenditures [...] a noticeable increase in 1974, followed by a **dip** in 1976.”

GPT can identify **Chart Junk Presence**

“The chart uses a highly decorative and non-essential illustration of a **monster** to frame the data. [...] This is known as ‘**chart junk**’ — visual elements that do not improve the reader's understanding.”

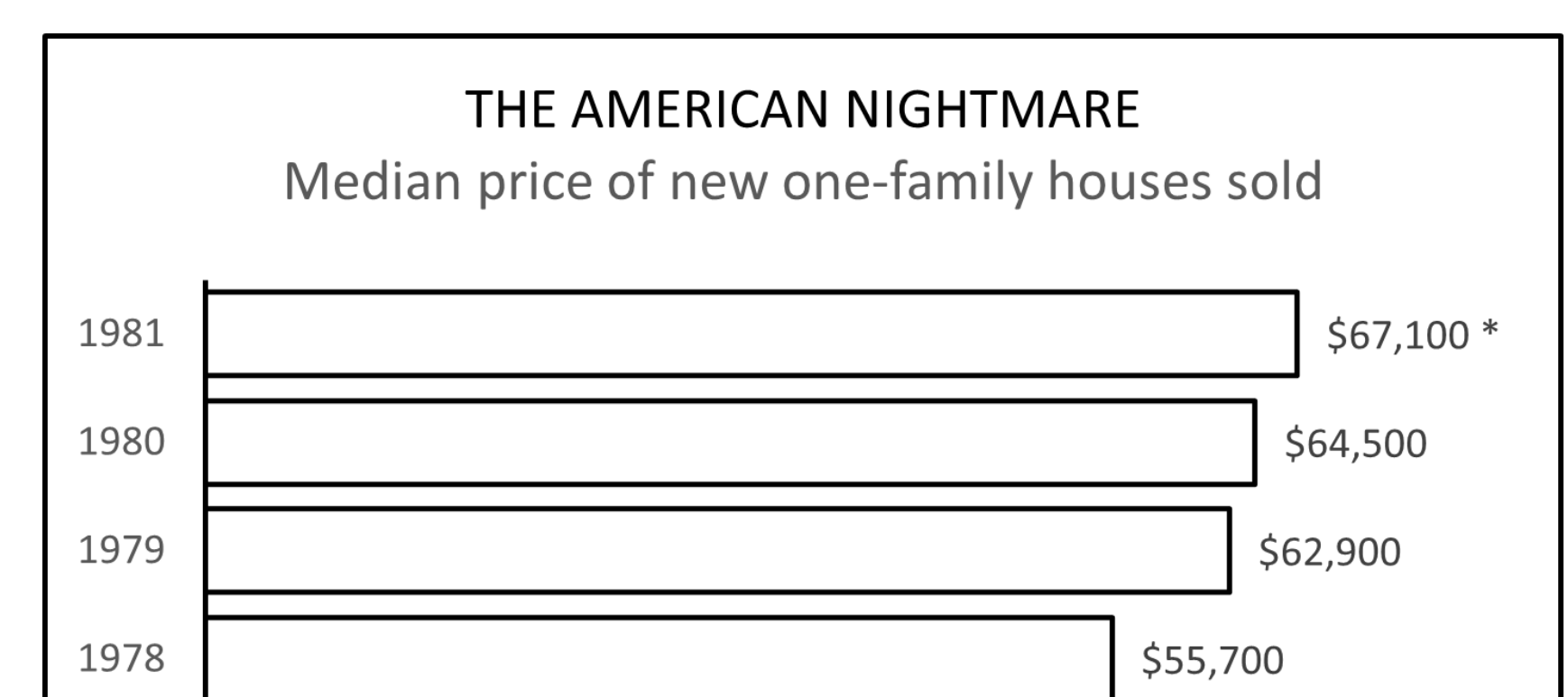


Plain Charts



“This chart contains **chart junk**. [...] **Pie charts themselves** are often considered chart junk. [...] The chart could be made clearer and more effective by using a bar chart.”

GPT has an expansive definition of “chart junk”



“The chart uses a **dramatic** and emotionally charged **title**, ‘THE AMERICAN NIGHTMARE’, which is **not necessary** for understanding the data and introduces bias.”

Key Takeaways

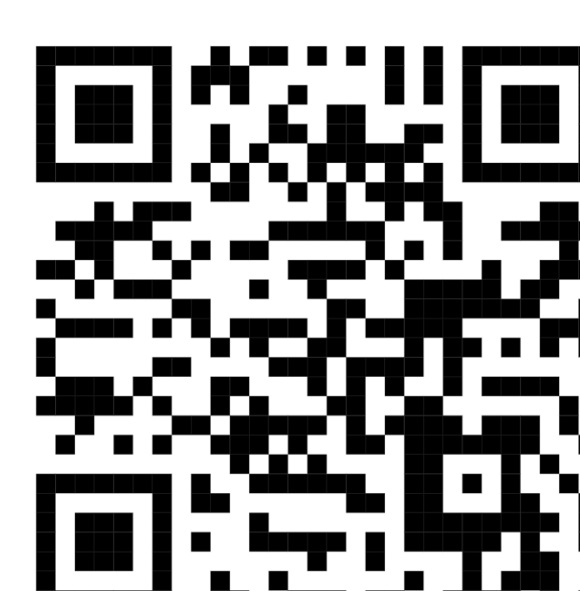
- **GPT-4.1 is generally successful** at reading visualizations with chart junk
- However, **chart junk could exacerbate existing challenges** at extracting exact values, especially when axes are impacted by junk
- **GPT-4.1 has a very broad definition of chart junk**, even for plain charts, placing many design aspects – and even entire chart types – in that category

Limitations & Next Steps

- We only test a single model, GPT-4.1
 - Test **open-source** models (e.g., LLaMA) & **reasoning** models (e.g., GPT-o4-mini)
- It is possible that the charts we use are in GPT’s training corpus
 - Evaluate using **new charts** that we can guarantee GPT has not seen before
 - A preliminary test on an original chart still showed ability to detect chart junk

References

- [1] A. Bendeck and J. Stasko. An Empirical Evaluation of the GPT-4 Multimodal Language Model on Visualization Literacy Tasks. *IEEE Trans. Vis. Comput. Graph.*, 31(1):1105–1115, 2025.
- [2] S. Bateman, R. L. Mandryk, *et al.* Useful Junk? The Effects of Visual Embellishment on Comprehension and Memorability of Charts. In *Proc. CHI*, pp. 2573–2582. ACM, 2010.
- [3] U. H. Syeda, L. South, *et al.* Towards a Deeper Understanding of Replication Studies: Replicating the Effects of Visual Embellishments on Comprehension and Memorability of Charts. *OSF Preprints*, 2024.



Check out the study materials!

